

Abdul Khurram

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Available for Co-ops/Internships (Open to Relocation): May–Dec 2026 | Full-time: May 2027+

EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Software Engineer Intern | *Python, TypeScript/Node.js, AWS, OpenAI*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Integrated **OpenAI Responses API** into **production** TypeScript/Node.js on **AWS**; built **agentic** content-generation **API services** serving **thousands/day** with **model inference** observability via CloudWatch.
- Enabled **AI-assisted** content creation by ingesting brand docs via an **NLP pipeline**, caching **embeddings**, and auto-injecting context while enforcing responsible **AI guardrails**.
- Strengthened **ML model reliability** to 99.9% via API migration, **cost/latency tracking**, retry/backoff, and **automated tests** in **CI/CD**; deployed via **Docker** containers.

Software Developer Co-op | *Python, Apache Spark, Airflow, SQL, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Engineered **Python/Airflow batch processing** pipelines handling **50M+ daily transactions** from Delta Lake, enabling downstream **ML/AI teams** with analysis-ready data on **petabyte-scale** systems.
- Optimized **Spark** jobs by migrating to Spark-native **APIs** and tuning AQE/cluster memory, cutting latency 40% and **compute costs** via **resource profiling**.
- Improving **batch processing** framework and **workflow orchestration** with **observability**, **fault tolerance**, and automated failover; instrumenting **cost/performance metrics** for **real-time** infra decisions.

RESEARCH EXPERIENCE

Software Developer & Team Lead | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led a human-AI collaboration research project: built an **AI-driven** VR simulation in Unreal (C++) pairing nurse practitioners with **AI patients** for clinical training.
- Designed a multithreaded supervisory architecture (C++/**Python**) enabling **real-time AI model** monitoring and protocol enforcement across agentic patient actors.
- Analyzed AI system performance, achieving 40% latency reduction and **90% inference cost savings** through **pipeline profiling** and **model optimization**.

PROJECTS

SoloPilot | *Python, TypeScript/Next.js, AWS, LangChain*

Personal Project

- Prototyping an AI-powered freelancer workspace with **agentic** workflows automating intake, scope analysis, proposal generation, and **deployment**.
- Designed an autonomous evaluator→revisor **agent loop** (**LangChain/Python**) with rubric-based benchmarking, improving quality through measured **model performance** analysis.
- Shipped an email intake console rendering versioned proposal PDFs via a **Vision Model**, demonstrating **production-level AI integration** into day-to-day workflows.

Backer | *TypeScript/Next.js, AWS DynamoDB, NextAuth*

Hack The Coast 2026

- Architected an AI-powered startup discovery platform (TypeScript/Next.js, DynamoDB) serving personalized investor feeds at **scale** via **API services**.
- Built **ML training infrastructure**: supervised **pipeline** with **sequence models** and **clustering** features, ingesting signals from DynamoDB event streams to train and **deploy** an **XGBoost** ranker to **production**.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript/JavaScript, C/C++, HTML/CSS

ML & Platforms: OpenAI API, LangChain, NLP/embeddings, Scikit-learn, PyTorch; Airflow, Apache Spark; AWS (Lambda, DynamoDB, S3, SQS, CloudWatch); Docker, Git, CI/CD